

Your grade: 100%

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1. Which of the following statements is true?

1 / 1 point

- ☐ A recurrence relation is an equation that defines a sequence based on a rule that gives the next term as a product of the previous terms
- ☒ Linear recurrence is a relation where each term of a sequence is a linear function of earlier terms in the sequence
- ☐ An infinite sequence is a function from the set of negative integers to the set of real numbers

✔ Nice work

Correct. This is the definition of a linear recurrence.

2. What is the object of the Hanoi tower game?

1 / 1 point

- ☒ The object of the game is to transfer all of the disks from spoke A to spoke C by moving one disk at a time without placing a larger disk on top of a smaller one
- ☐ The object of the game is to transfer all of the disks from spoke A to spoke C by moving one disk at a time without placing a smaller disk on top of a larger one
- ☐ The object of the game is to transfer all of the disks from spoke A to spoke B by moving one disk at a time without placing a larger disk on top of a smaller one

✔ Nice work

Correct. This is the objective of the Hanoi tower game.

3. Which of the following is a type of linear recurrence?

1 / 1 point

- ☒ Linear non-homogeneous recurrences
- ☐ Linear non-heterogenous recurrences
- ☐ Linear heterogenous recurrences

✔ Nice work

Correct. Linear non-homogeneous recurrences are a type of linear recurrence.

4. Let's consider the following case:

1 / 1 point

- A country with currently 70 million people
- A population growth rate (birth rate minus death rate) of 2% per year
- Receives 50, 000 immigrants per year
- Every year 30, 000 people emigrate from the country

Which of the following formalises the population in year n as linear recurrence a_n ?

- ☐ $a_0 = 70000000, \forall n \in \mathbb{N} \ a_{n+1} = 2.a_n + 20000$
- ☐ $a_0 = 70000000, \forall n \in \mathbb{N} \ a_{n+1} = 1.02a_n + 50000$
- ☒ $a_0 = 70000000, \forall n \in \mathbb{N} \ a_{n+1} = 1.02a_n + 20000$

✔ Nice work

Correct. This correctly incorporates the population growth rate and net immigration.

5. Which of the following statements is true?

1 / 1 point

- ☒ A sequence is called geometric if the ratio between consecutive terms is a constant r
- ☐ A sequence is called arithmetic if the ratio between consecutive terms is a constant c
- ☐ A divide-and-conquer algorithm consists of merging the problems into one problem and solving it

✔ Nice work

Correct. This is the definition of a geometric sequence.